

~\Desktop\LAB\cqueue.c

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1 #include<stdio.h>
2 int A[20],size,front,rear;
3 // Function for inserting an item to Circular Queue
4 void ENQUEUE_CQ(int item)
5 {
6     if(((rear+1)%size)==front)
7         printf("Queue is Full ,insertion is not possible\n");
8     else if(rear== -1)
9     {
10         front=0;
11         rear=0;
12         A[rear]=item;
13     }
14     else
15     {
16         rear=(rear+1)%size;
17         A[rear]=item;
18     }
19 }
20 // Function for deleting an item from Circular Queue
21 void DEQUEUE_CQ()
22 {
23     if(front<= -1)
24         printf("Queue is empty\n");
25     else if(front==rear)
26     {
27         printf("Deleted item is %d\n",A[front]);
28         front=-1;
29         rear=-1;
30     }
31     else
32     {
33         printf("Deleted item is %d \n", A[front]);
34         front=(front + 1)%size;
35     }
36 }
37 // Function for displaying the items in the Circular Queue
38 void DISPLAY()
39 {
40     int i;
41     if(front== -1)
42         printf("Queue is empty\n");
43     else if(front<=rear)
44     {
45         printf("f->");
46         for(i=front;i<=rear;i++)
47         {
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48             printf("%d\t",A[i]);
49         }
50         printf("<-r\n");
51         printf("front=%d\n", front);
52         printf("rear=%d\n", rear);
53     }
54     else
55     {
56         printf("f->");
57         for(i=front;i<=size-1;i++)
58             printf("%d\t",A[i]);
59         for(i=0;i<=rear;i++)
60             printf("%d \t",A[i]);
61         printf("<-r\n");
62         printf("front=%d\n", front);
63         printf("rear=%d\n", rear);
64     }
65 }
66 // Function for main()
67 int main()
68 {
69     int opt,item;
70     rear=-1;
71     front=-1;
72     printf("Enter the size of circular queue: \n");
73     scanf("%d",&size);
74     do
75     {
76         printf("Menu\n");
77         printf("\n 1.ENQUEUE\t 2.DEQUEUE \t 3.DISPLAY \t 4.EXIT");
78         printf("\n Enter the option:");
79         scanf("%d",&opt);
80         switch(opt)
81         {
82             case 1 : printf("Enter the item to be inserted: \n");
83                       scanf("%d",&item);
84                       ENQUEUE_CQ(item);
85                       DISPLAY();
86                       break;
87             case 2 : DEQUEUE_CQ();
88                       DISPLAY();
89                       break;
90             case 3 : DISPLAY();
91                       break;
92             case 4 : break;
93
94             case 5 : printf(" \nInvaild choice!Please try again");
95         }
96     }while(opt!=4);
97 }
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98 OUTPUT
99 Enter the size of circular queue:
100 3
101 Menu
102 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT
103 Enter the option:2
104 Queue is empty
105 Menu
106 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT
107 Enter the option:1
108 Enter the item to be inserted:
109 5
110 f->5 <-r
111 front=0
112 rear=0
113 Menu
114 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT
115 Enter the option:1
116 Enter the item to be inserted:
117 5
118 f->5 5 <-r
119 front=0
120 rear=1
121 Menu
122 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT
123 Enter the option:1
124 Enter the item to be inserted:
125 9
126 f->5 5 9 <-r
127 front=0
128 rear=2
129 Menu
130 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT
131 Enter the option:
132 1
133 Enter the item to be inserted:
134 9
135 Queue is Full ,insertion is not possible
136 f->5 5 9 <-r
137 front=0
138 rear=2
139 Menu
140 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT
141 Enter the option:2
142 Deleted item is 5
143 f->5 9 <-r
144 front=1
145 rear=2
146 Menu
147 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT

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148 Enter the option:2
149 Deleted item is 5
150 f->9 <-r
151 front=2
152 rear=2
153 Menu
154 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT
155 Enter the option:1
156 Enter the item to be inserted:
157 6
158 f->9 6 <-r
159 front=2
160 rear=0
161 Menu
162 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT
163 Enter the option:
164 1
165 Enter the item to be inserted:
166 9
167 f->9 6 9 <-r
168 front=2
169 rear=1
170 Menu
171 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT
172 Enter the option:2
173 Deleted item is 9
174 f->6 9 <-r
175 front=0
176 rear=1
177 Menu
178 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT
179 Enter the option:2
180 Deleted item is 6
181 f->9 <-r
182 front=1
183 rear=1
184 Menu
185 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT
186 Enter the option:2
187 Deleted item is 9
188 Queue is empty
189 Menu
190 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT
191 Enter the option:4
192
193
194
195
196
197

```